

# Metabolic vitamin B12 status on a mostly raw vegan diet with follow-up using tablets, nutritional yeast, or probiotic supplements.

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**BACKGROUND:** Pure vegetarian diets might cause cobalamin deficiency due to lack of dietary intake. It was hypothesized that a population following a vegan diet consuming mostly raw fruits and vegetables, carrot juice, and dehydrated barley grass juice would be able to avoid vitamin B12 deficiency naturally. **METHODS:** Subjects were recruited at a health ministers' reunion based on adherence to the Hallelujah diet for at least 2 years. Serum cobalamin and urinary methylmalonic acid (MMA) assays were performed. Follow-up with sublingual tablets, nutritional yeast, or probiotic supplements was carried out on subjects with abnormal MMA results. **RESULTS:** 49 subjects were tested. Most subjects (10th to 90th percentile) had followed this diet 23-49 months. 6 subjects had serum B12 concentrations <147 pmol/l (200 pg/ml). 37 subjects (76%) had serum B12 concentrations <221 pmol/l (300 pg/ml). 23 subjects (47%) had abnormal urinary MMA concentrations above or equal to 4.0 microg/mg creatinine. Sublingual cyanocobalamin and nutritional yeast, but not probiotic supplements, significantly reduced group mean MMA concentrations (tablet  $p < 0.01$ ; yeast  $p < 0.05$ , probiotic  $> 0.20$ ). **CONCLUSIONS:** The urinary MMA assay is effective for identifying early metabolic cobalamin deficiency. People following the Hallelujah diet and other raw-food vegetarian diets should regularly monitor their urinary MMA levels, consume a sublingual cobalamin supplement, or consume cobalamin in their food.